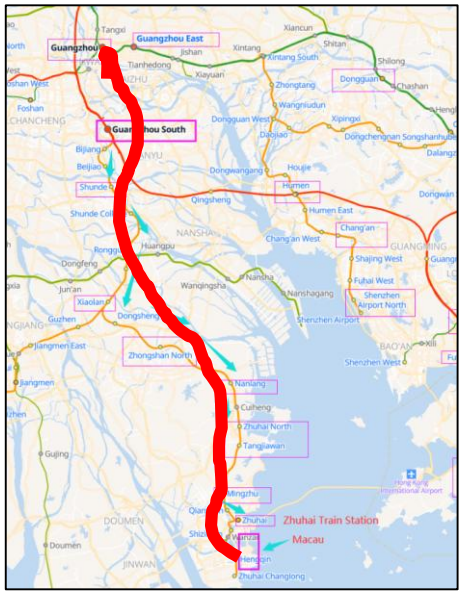
Deep Learning and its Applications to Pollen Analysis

Shu Kong

University of Macau
https://aimerykong.github.io/
skong@um.edu.mo

43rd Mid-Continent Paleobotanical Colloquium
Panama 2026

What lead me here?



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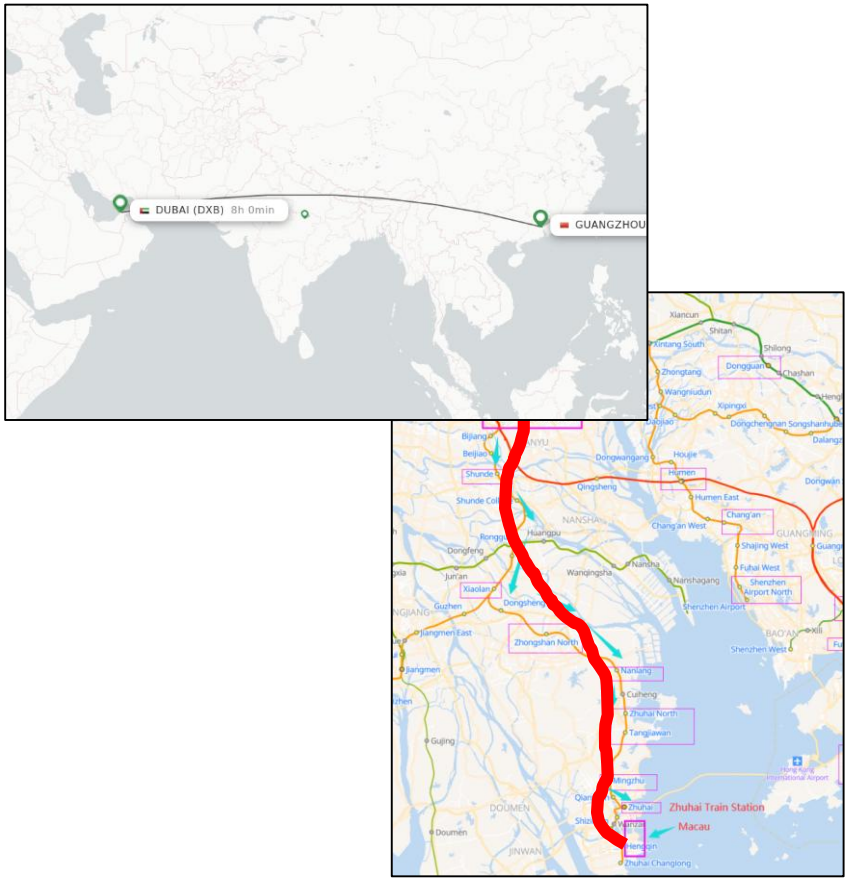

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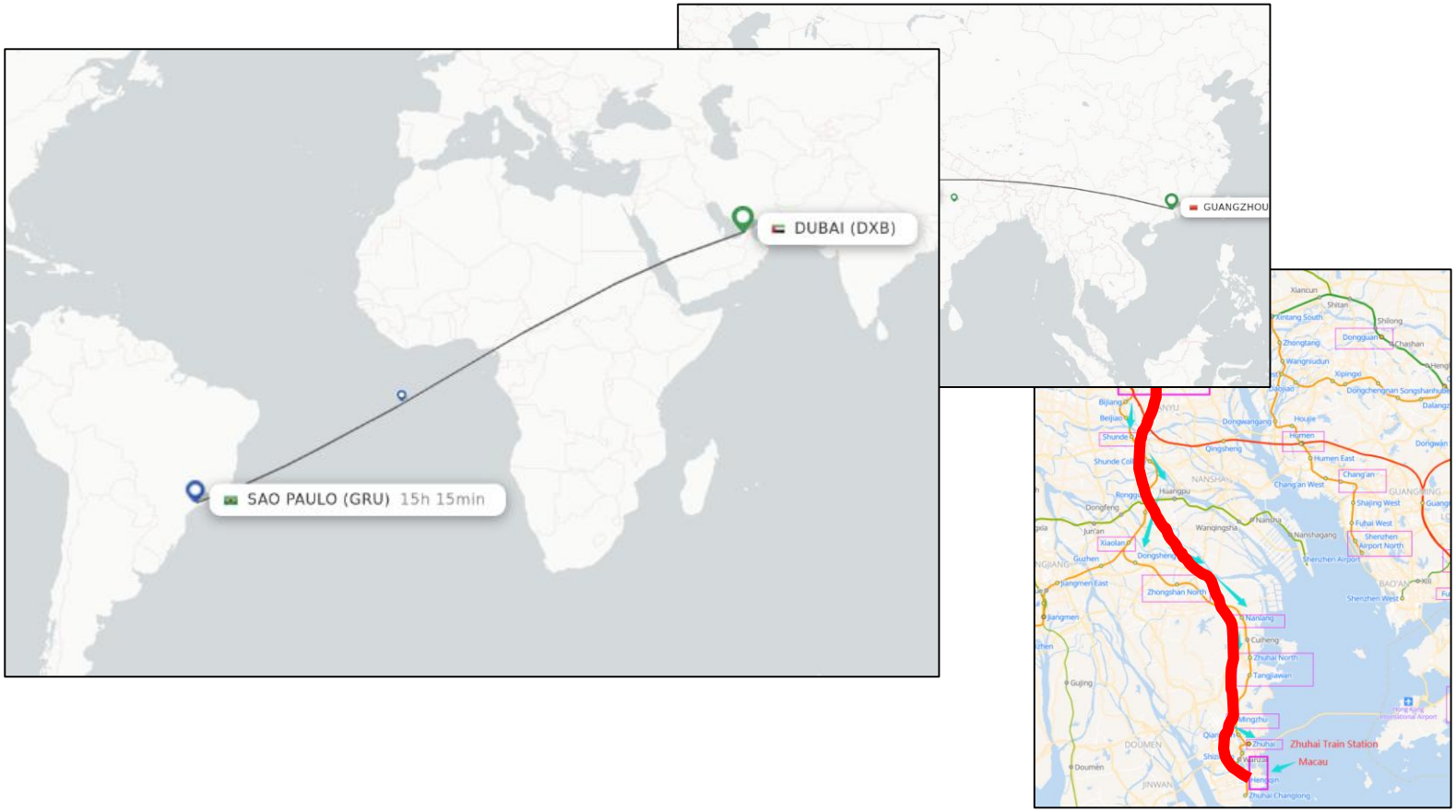

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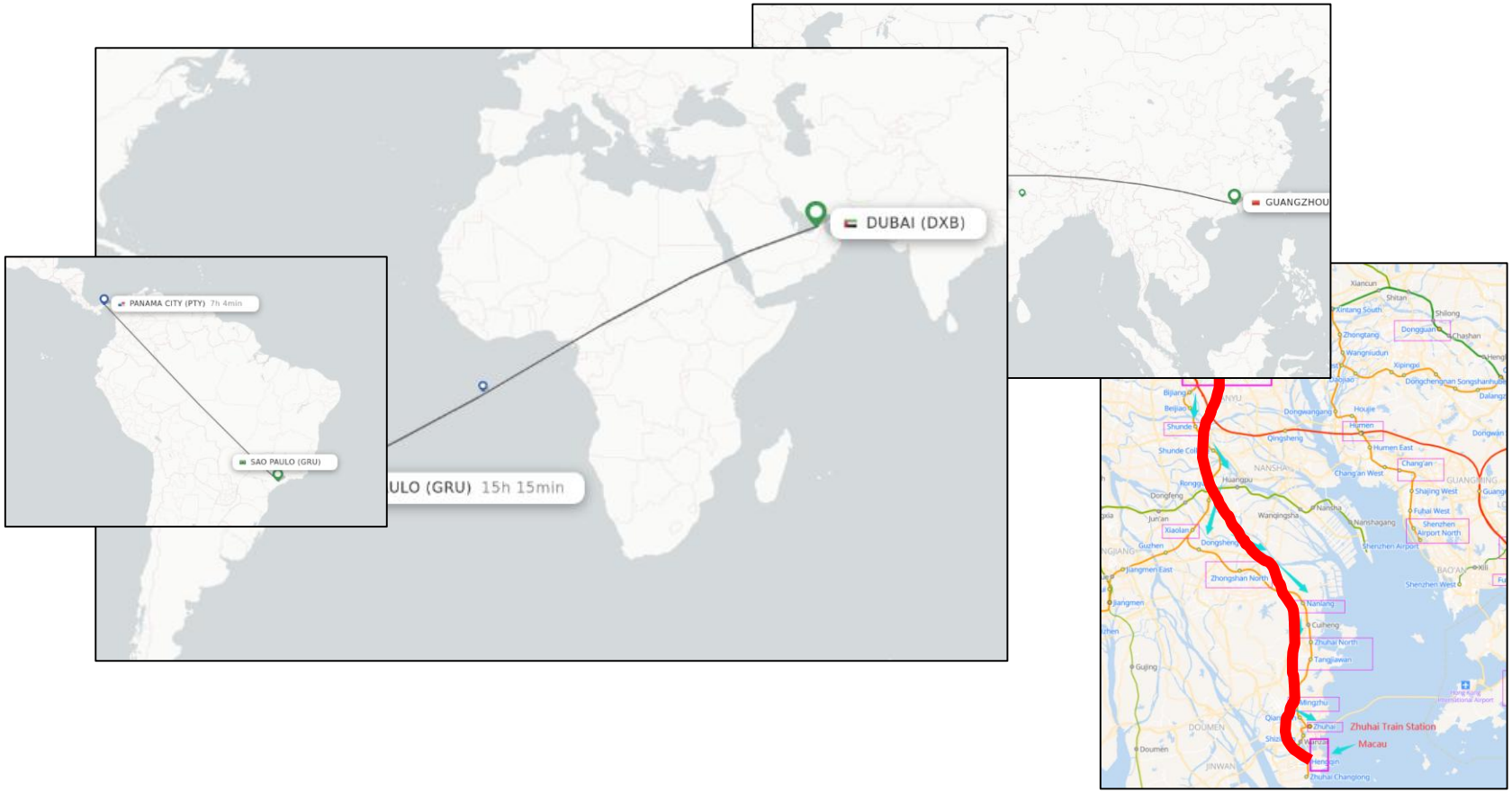
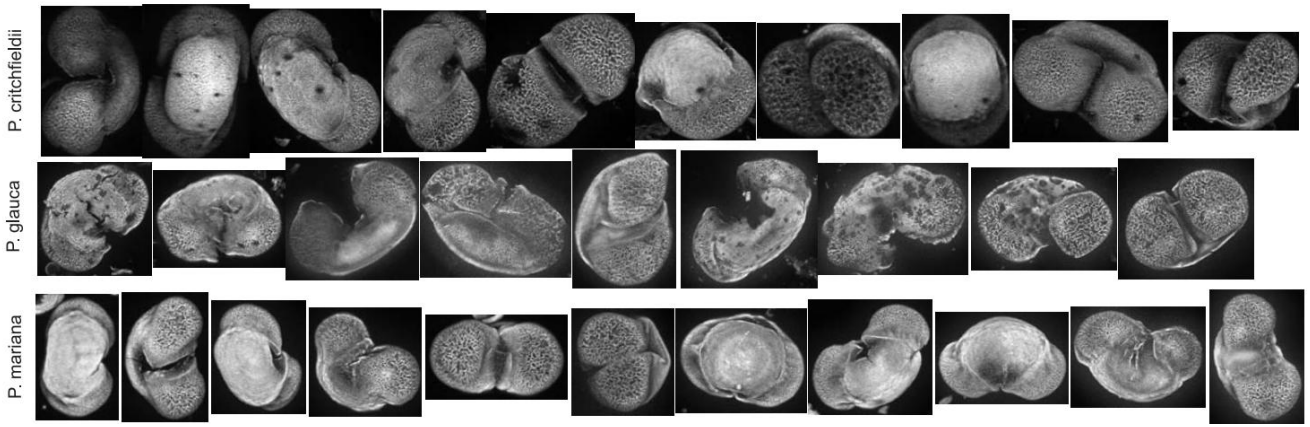
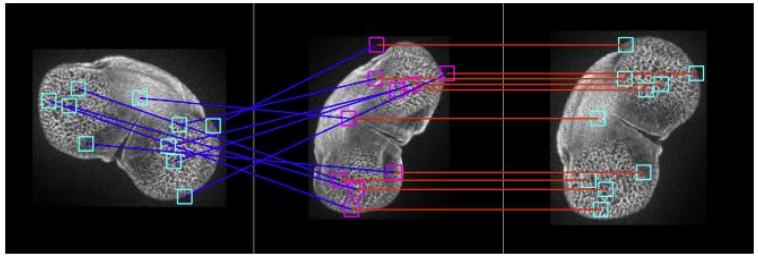


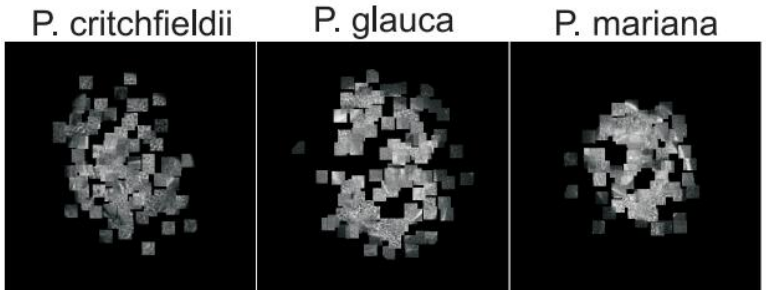
What lead me here?

2014 PhD

Non-deep learning for pollen recognition







Non-deep learning for fossil pollen recognition with Zeiss Apotome fluorescence microscope

S. Kong, S. Punyasena, C. Fowlkes, "Spatially Aware Dictionary Learning and Coding for Fossil Pollen Identification", CVPRW, 2016

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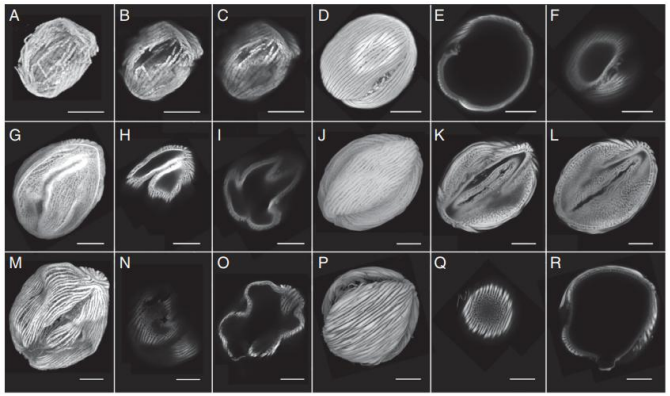

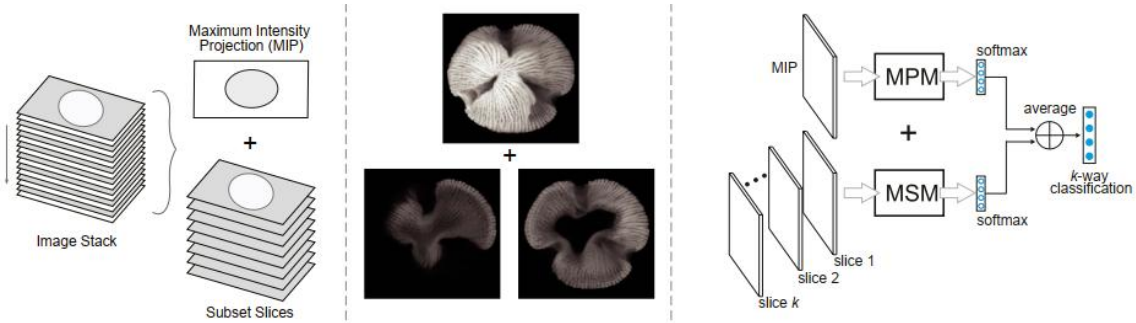

Deep learning for fossil pollen recognition with Airyscan confocal superresolution microscopy

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What lead me here?

2014 PhD — Postdoc

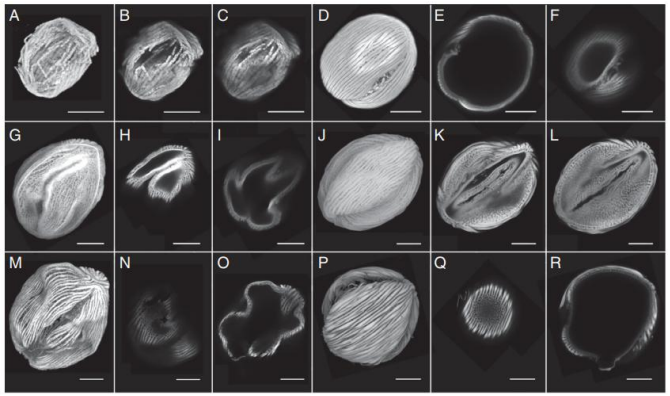
Non-deep learning for pollen recognition

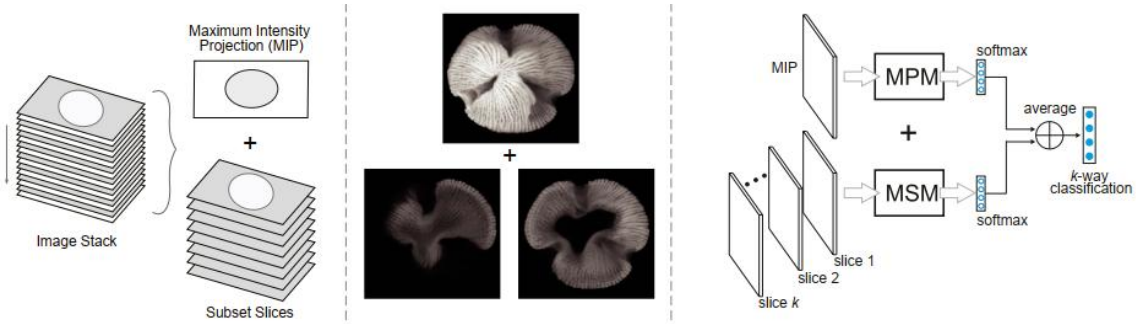
Deep learning for pollen recognition

NSF Stories

Modern computational tools open new era of fossil pollen research

Integrating machine-learning technology with high-resolution imaging helps identify plant species





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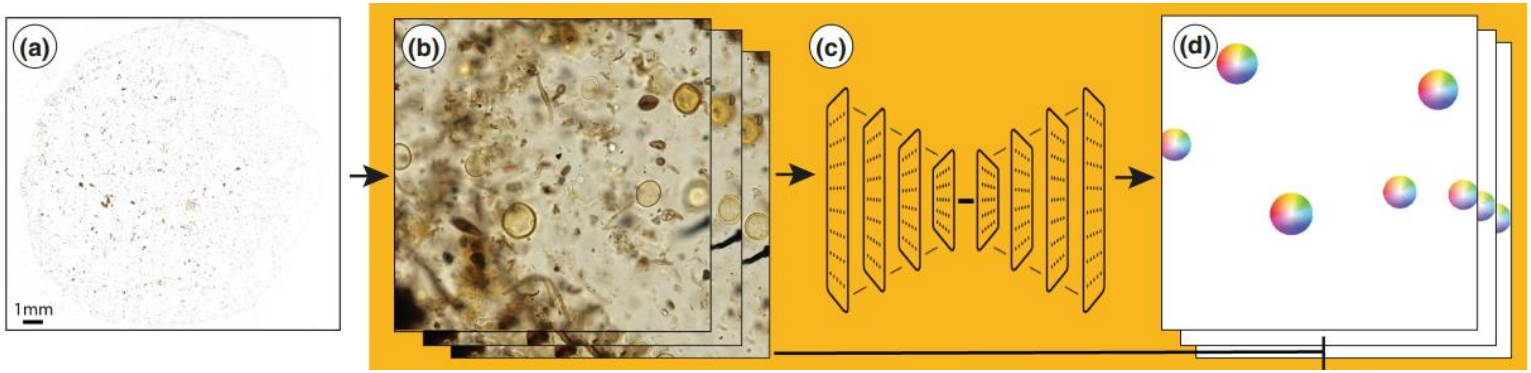
What lead me here?

2014 PhD — Postdoc — faculty

Non-deep learning for pollen recognition

Deep learning for pollen recognition

Deep learning for pollen detection



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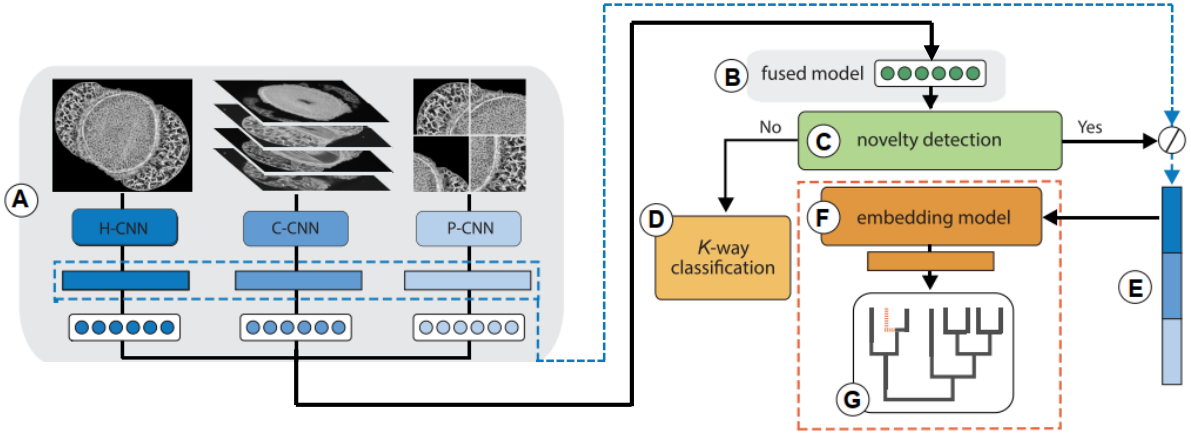
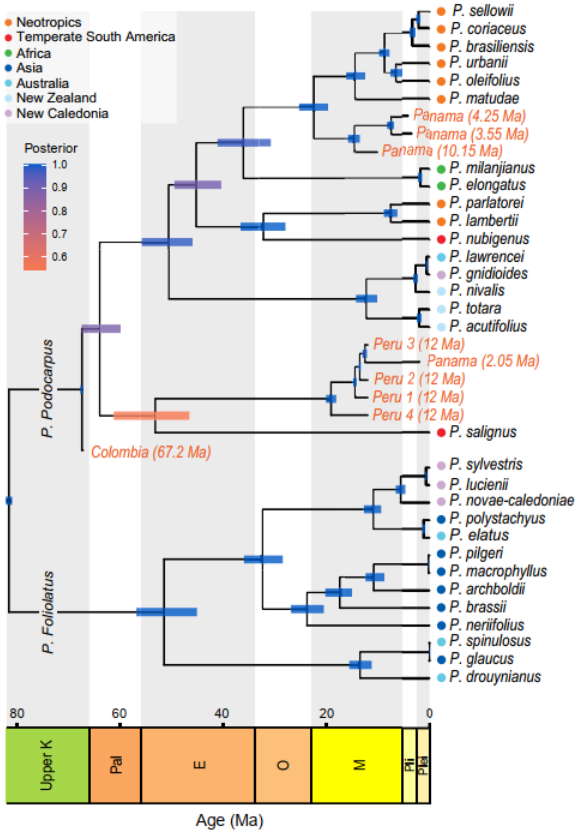
PhD	Postdoc	faculty	faculty
Non-deep learning for pollen recognition	Deep learning for pollen recognition	Deep learning for pollen detection	Deep learning for deep pollen analysis





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What lead me here?

1. Meet people.
2. Do something together.
3. Enjoy a short stay in Panama.

What to get from this short course?

1. *AI –related topics*: computer vision, machine learning, deep learning
2. *Hand-on experience*: image processing, model training, foundation model adaptation
3. A safe venue to discuss interdisciplinary research

Lecture	Topics
1. Introduction	machine learning, computer vision, deep learning, artificial intelligence
2. Basics via Classification	Nearest Neighbor, Perceptron, Logistic Regression, cross entropy, feature, gradient descent, etc.
3. Deep Neural Networks	Multi-Layer Perceptron, feature learning, backpropagation, CNN, Transformer
4. Hands-On Practice: Basics	pollen data preparation, visualization, nearest neighbor, linear classifier, CNN
5. Segmentation and Detection	semantic segmentation, instance segmentation, U-Net, pixel grouping, mean-shift
6. Foundation Models	CLIP, Segment Anything Model, DINOv2, GroundingDINO, zero-shot learning
7. Advanced Topics	open-set recognition, novel species recognition, phylogentic reconstruction via clustering
8. Hands-On Practice: Advanced Topics	finetuning, pollen detection/segmentation

What we will learn?

- Pollen Recognition
- Pollen Detection
- Novel Taxa Detection
- Phylogenetic Reconstruction

⟷

- Image Classification
- Object Detection
- Open-Set Recognition
- Clustering

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Roadmap

Toolboxes:

- platform: Python, Jupyter Notebook
- basics: sklearn, skimage, numpy, scipy, matplotlib
- for deep learning: PyTorch
- Teaching materials are available at https://github.com/aimerykong/MPC2026-tutorial

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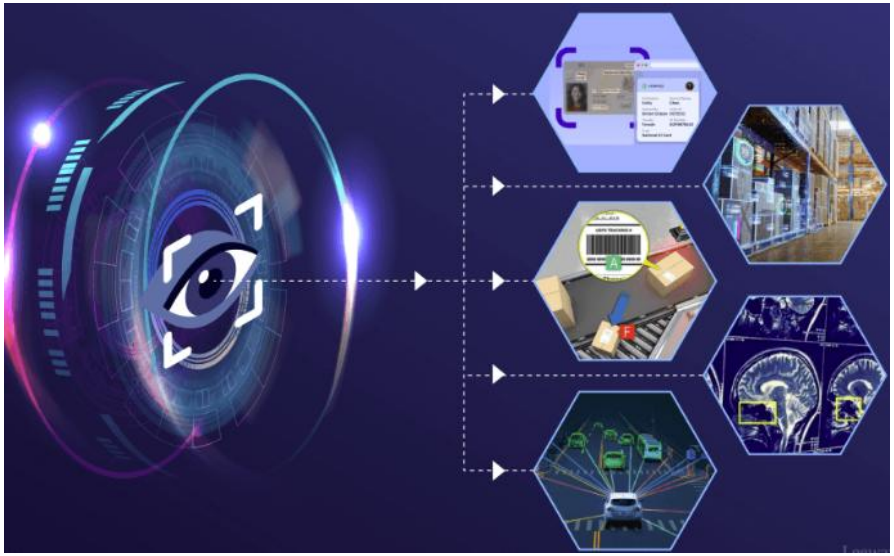
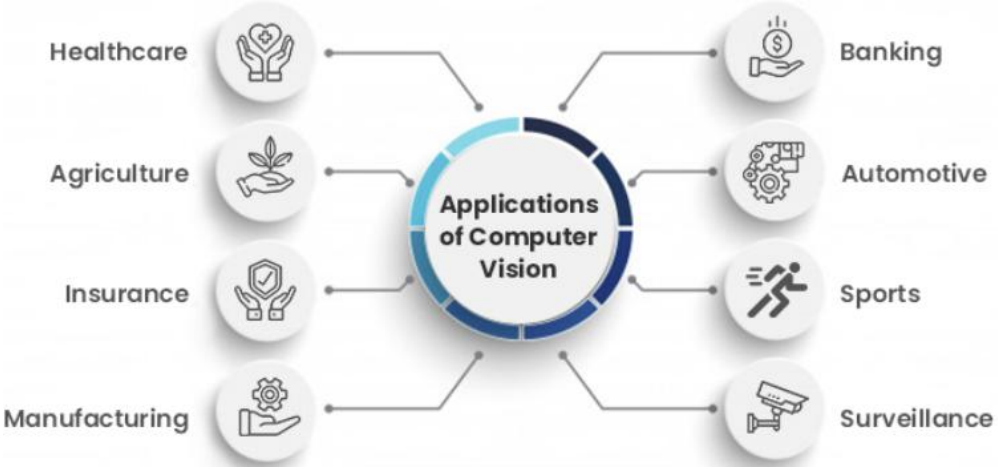
Terminology

Computer Vision (CV) is a field of artificial intelligence (AI) that enables computers to interpret, analyze, and understand *visual data* from the world, such as images and videos, to make decisions or recommendations.





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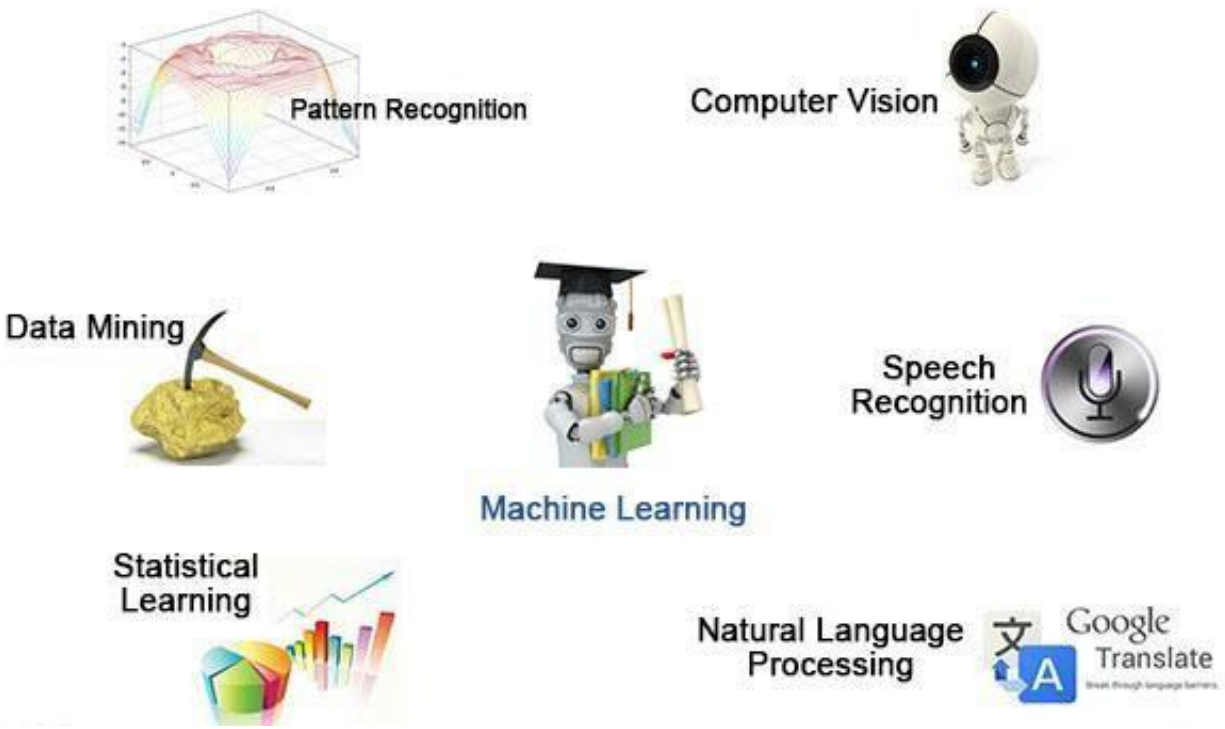

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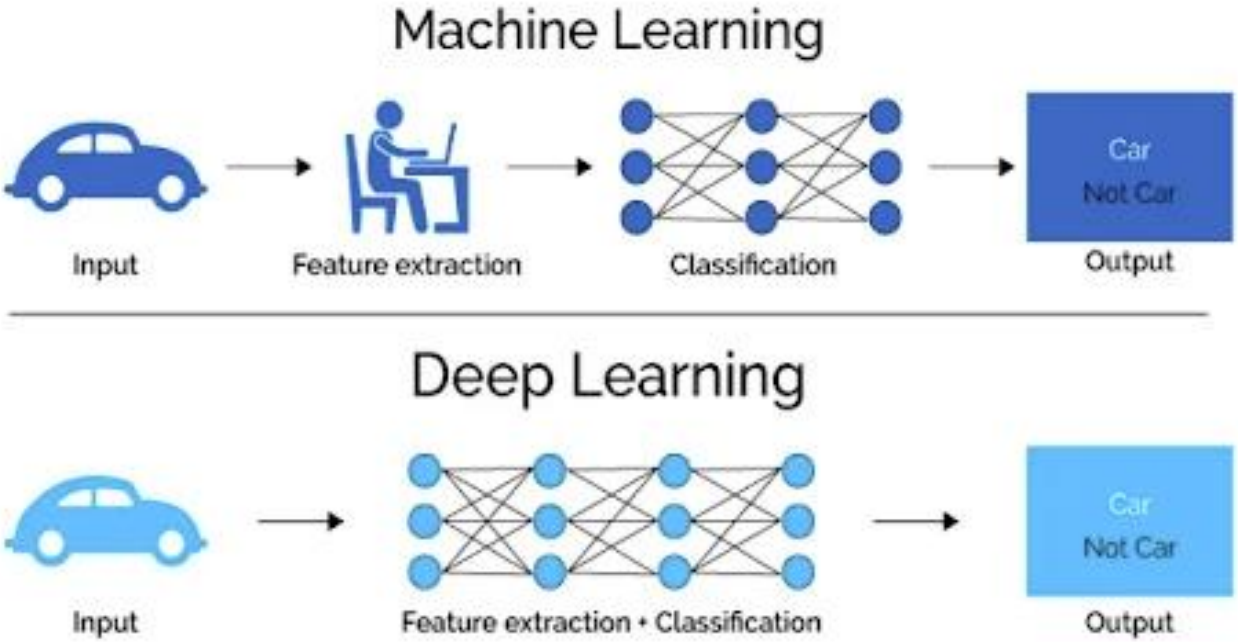

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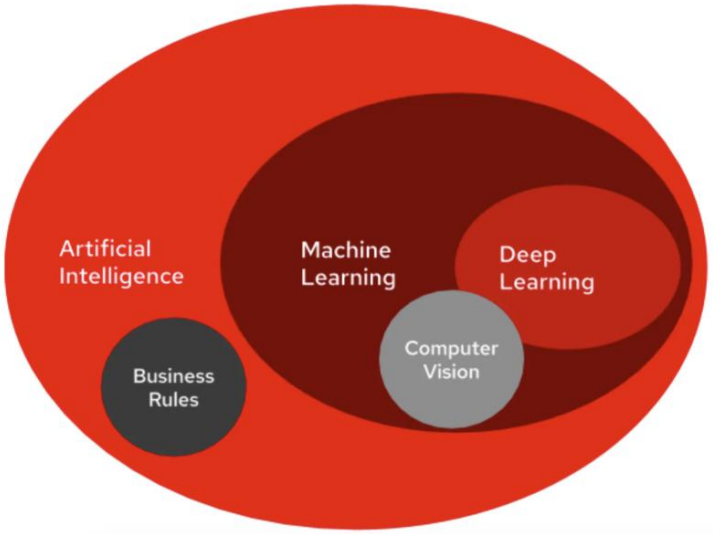
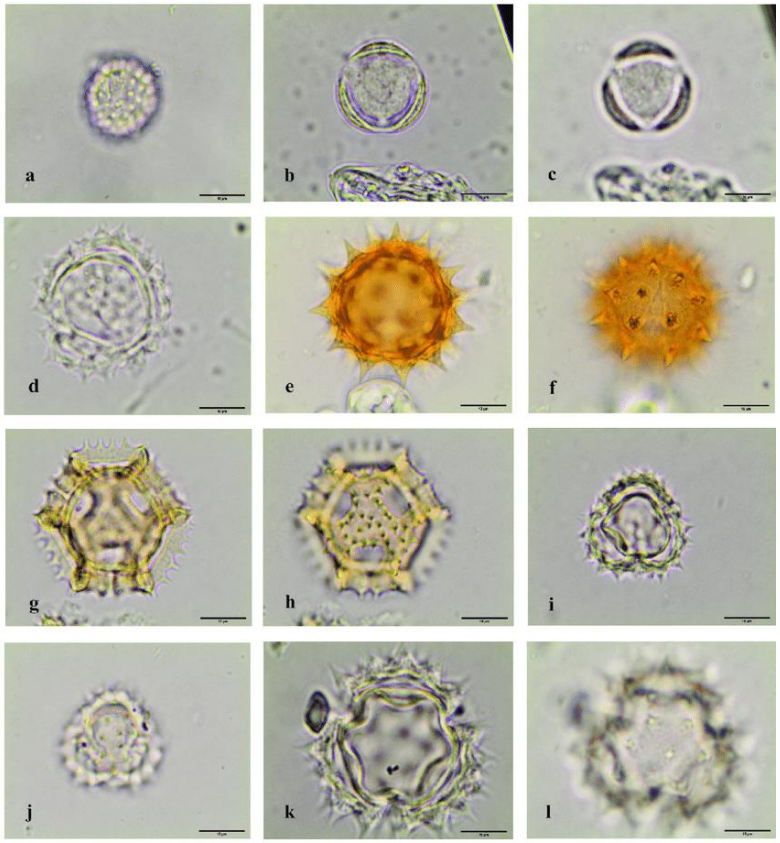
Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, particularly computer systems, enabling them to learn, reason, problem-solve, and perceive.



Pollen Recognition



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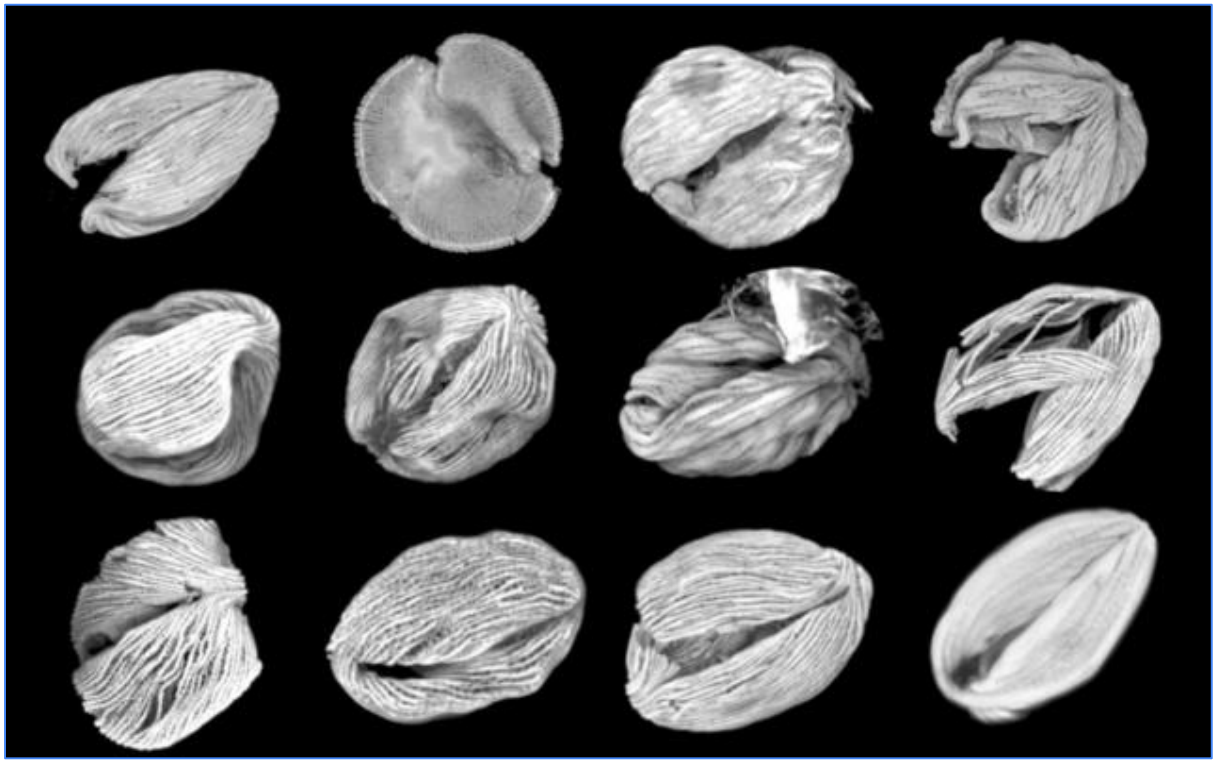

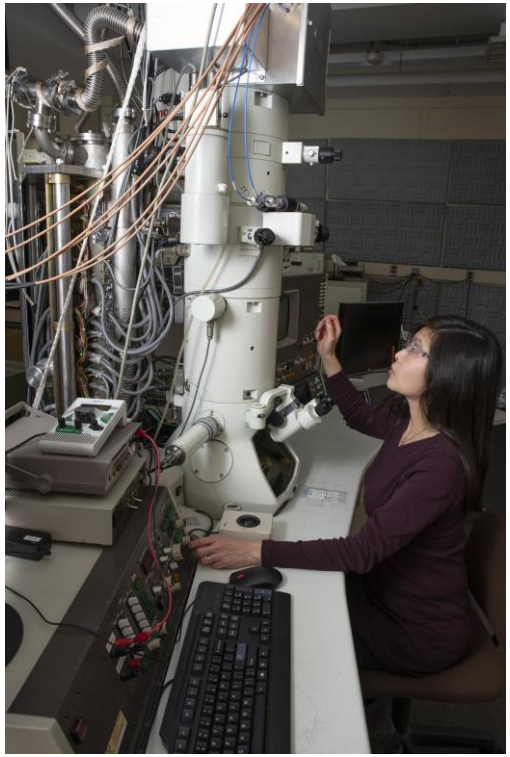

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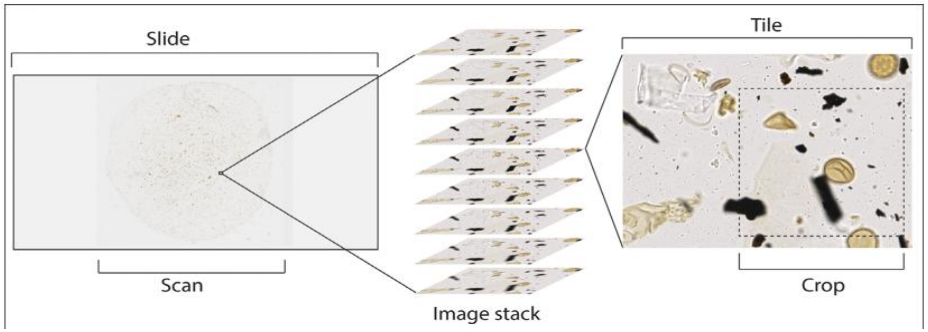
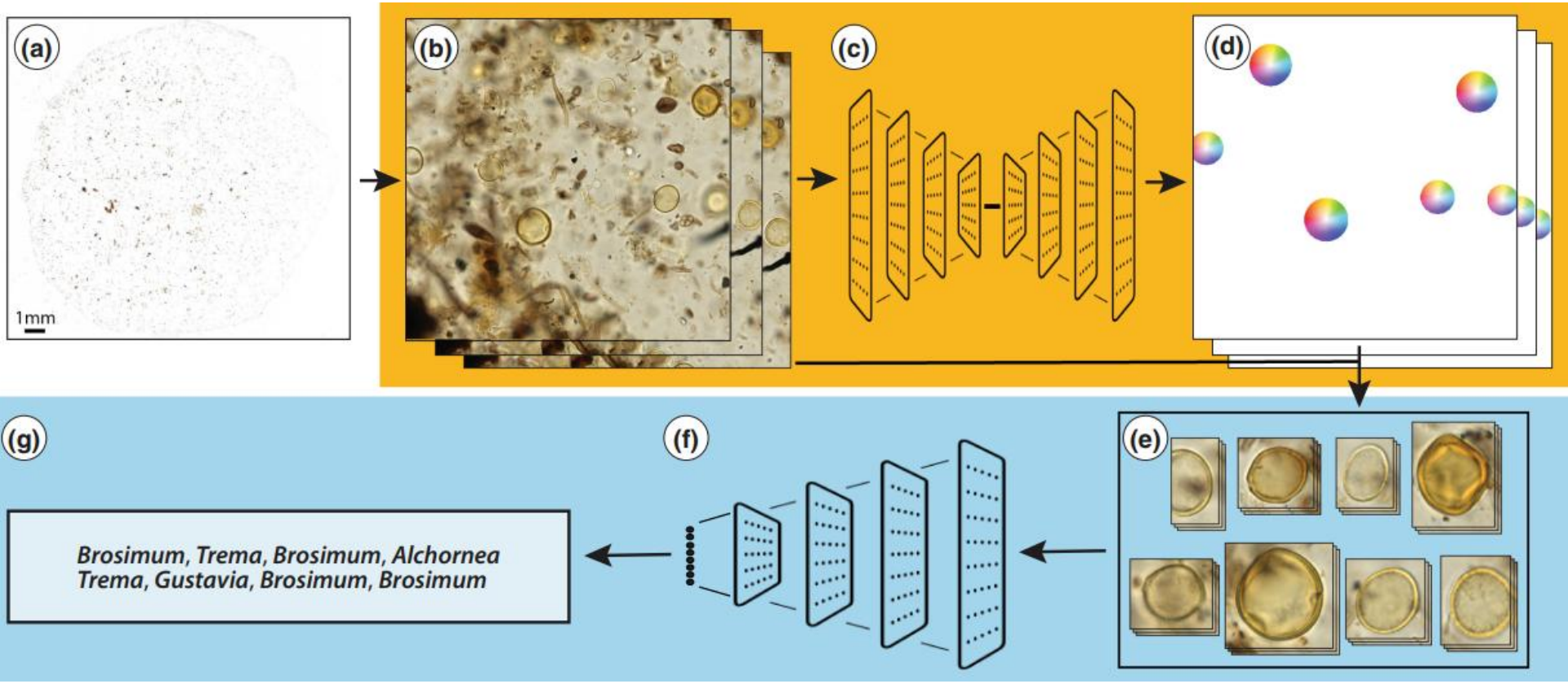
Pollen Detection



A Pipeline: Pollen Detection + Recognition



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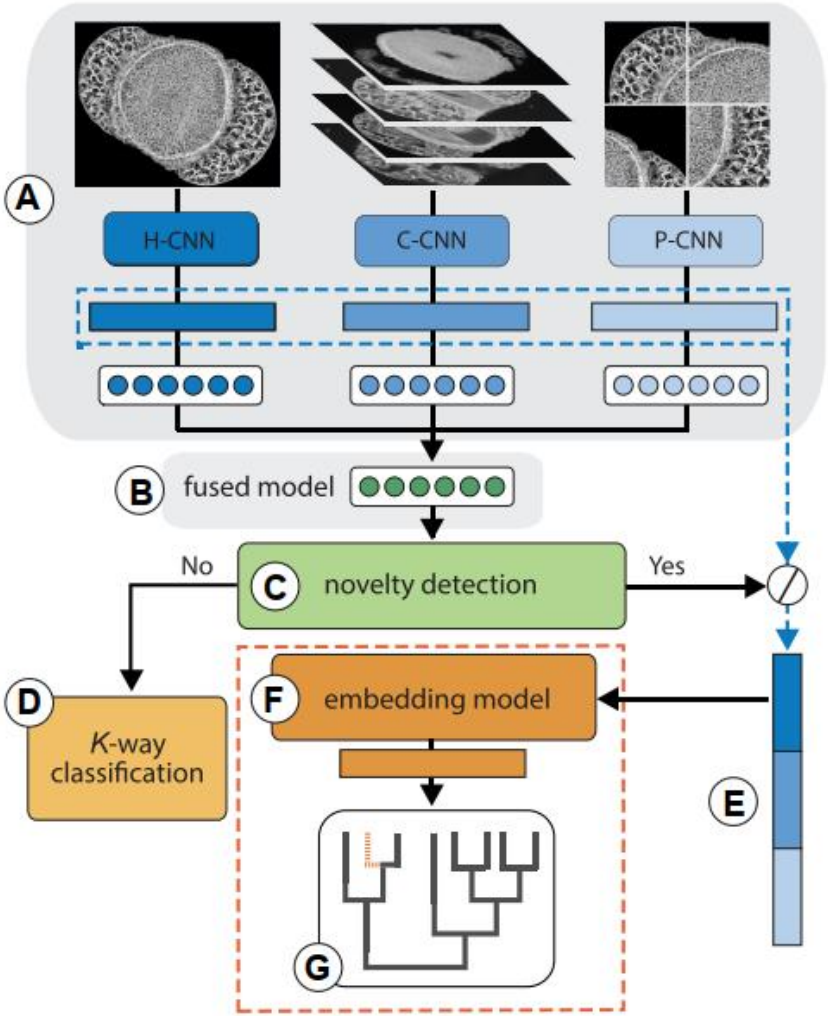
Beyond Recognition: Novel Taxa Detection



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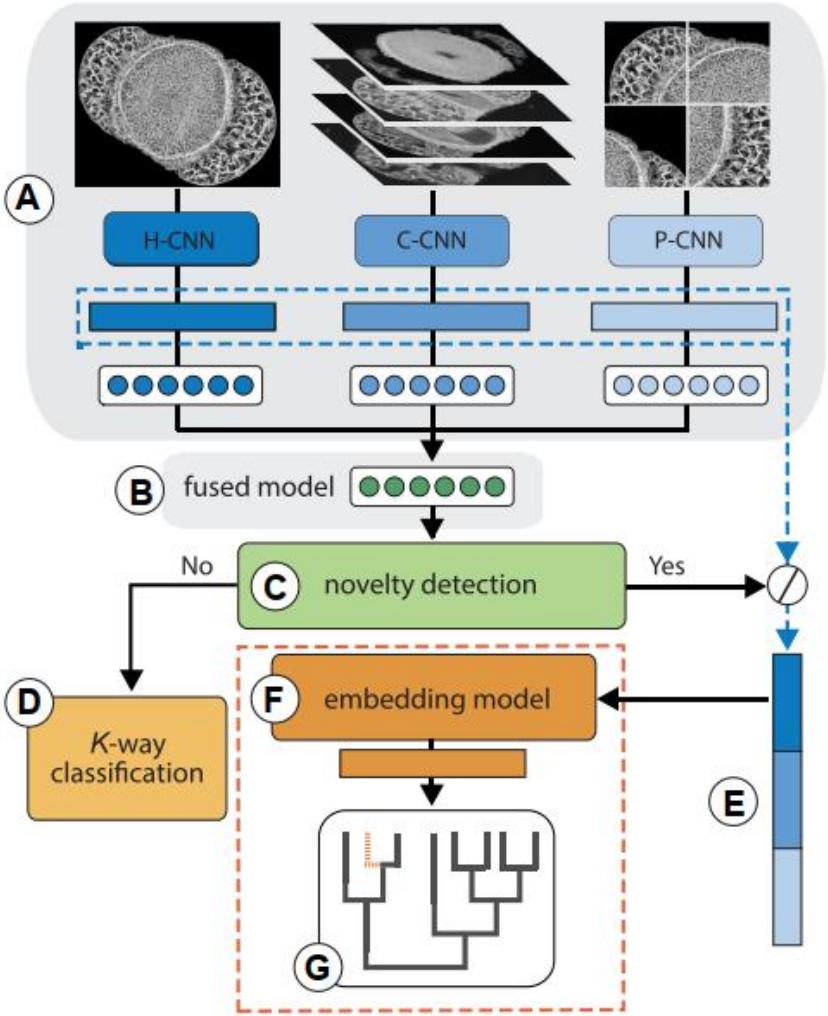
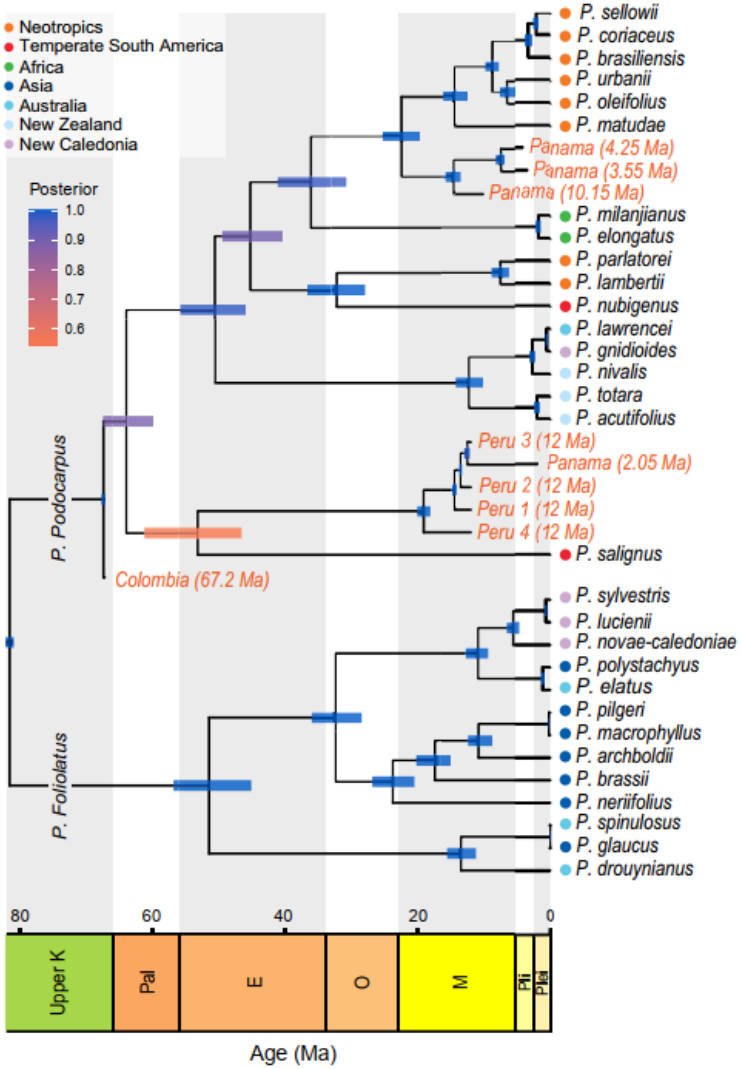
Beyond Recognition: Phylogenetic Reconstruction





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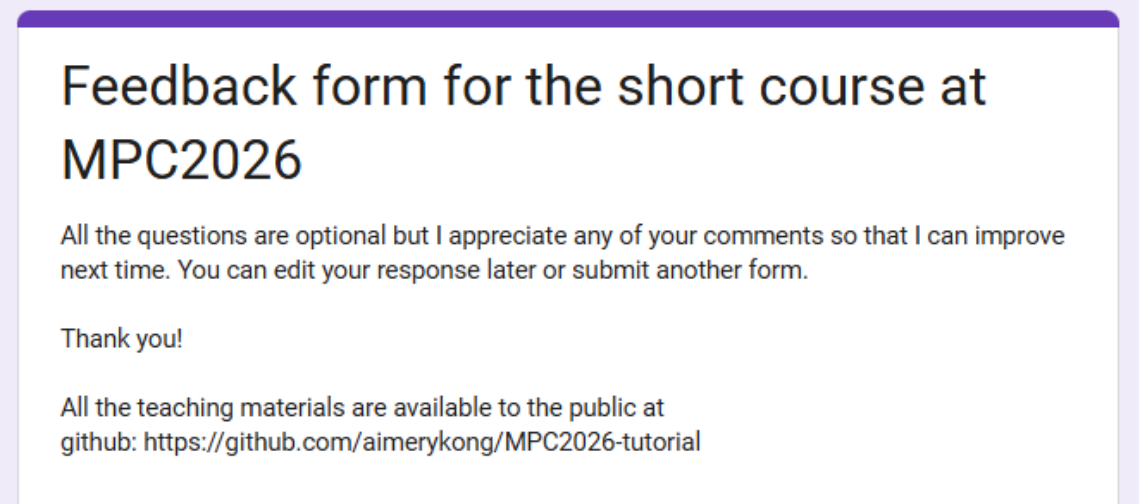
Misc

- Let’s make it interactive.
- Let’s make it comfy.
- Let’s make it continual.

- Feedback after this course: https://forms.gle/RpdMyRoEhCNiJYsM9



Feedback form for the short course at MPC2026

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Miscellaneous

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MPC2026-tutorial Public

main | 1 Branch | 0 Tags

aimerykong Rename code_000_readme.pdf to code

- dataset
- misc
- slides
- README.md
- code000_readme.pdf
- code001_setup.ipynb
- code002_data_prep_NearestNeighbor.ipynb
- code003_LogReg.ipynb
- code004_DNN_scratch.ipynb
- code005_DNN_finetune.ipynb
- code006_detection_dataPrep.ipynb
- code007_pollen_detection_model_design.ipynb
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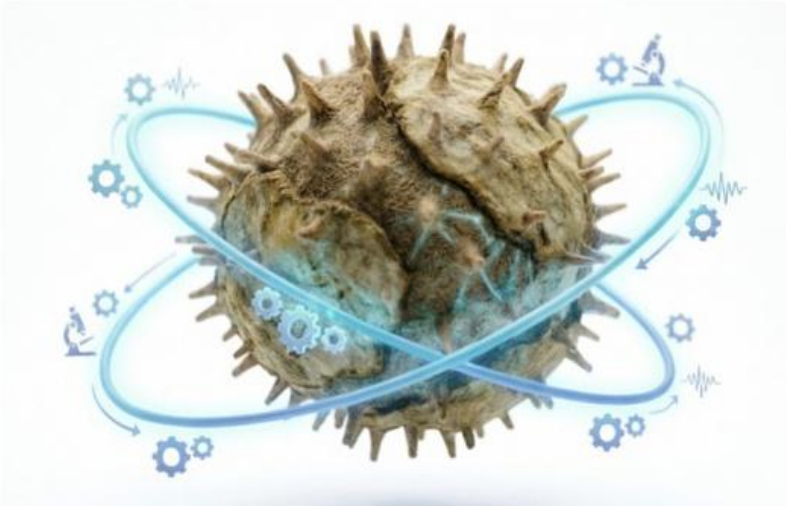
Short Course: Deep Learning and its Applications to Pollen Analysis

by Shu Kong



This repository hosts teaching materials for the short course at the 43rd Mid-Continent Paleobotanical Colloquium. Focusing on the theme of applying deep learning to automate pollen analysis, this course covers relevant techniques in computer vision, machine learning, and deep learning. It will provide hands-on coding experience to demonstrate how to apply these techniques to pollen image processing, recognition, segmentation/detection, novel species recognition, and so on. It will also teach how to exploit foundation models or AI agents for more precise pollen recognition and detection. Slides will be uploaded after this short course. Refer to the following for the syllabus and important dates.

Miscellaneous

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